

Katie de la Pena

Professor LeBlanc

COMP 255 - Artificial Intelligence

28 September 2026

Program 1: Algorithm Search Comparison Report

The purpose of this experiment was to analyze the average runtime of three different search algorithms in Python. Using various different input sizes (N), each algorithm was run with 3 trials, and 1000 times in each trial. To calculate the minimum runtime comparison time of the algorithms, the average search time for each trial was divided by the sample size (1000) to compute an estimated average single search time for each trial. The smallest value of the three trials was listed as the minimum time. As displayed in all figures below, Linear Search had the longest minimum runtimes as the input size (N) increased. When comparing the two different computers (Linux and Macbook Pro), the minimum runtime comparisons seem pretty similar except for the binary and interpolation algorithms, especially towards higher input values such as 10^6 and 10^7 , where Figures 2 and 4 show a slight time increase on the Linux box whereas Figures 1 and 3 show a bit of an increase, but a pretty flat slope on the Macbook Pro. Overall, Linear Search appears to be the least efficient out of all 3 algorithms based on the 4 graphs, and interpolation search was the most efficient.

Linear vs Binary Vs Interpolation Search MINIMUM Runtime Comparison on Macbook Pro

(Over 3 Trials, 1000 Samples)

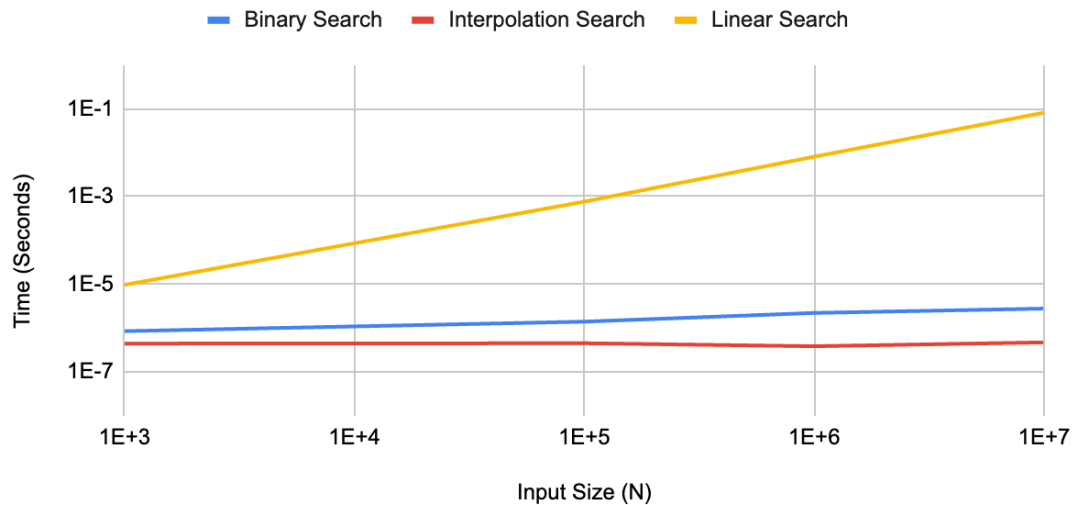


Figure 1. Graph created through Google Sheets using “macbookPro_search_results.csv”.

Displays minimum runtime comparisons across three different search algorithms given different input sizes

Macbook Pro Specs: Apple M3 Pro, 18 GB, MacOS Sequoia Version 15.6.1

Minimum Time calculated by calculating average trial times divided by sample size, and identifying smallest value

Linear vs Binary vs Interpolation Search MINIMUM Runtime Comparison on Linux Box

(over 3 trials, 1000 samples)

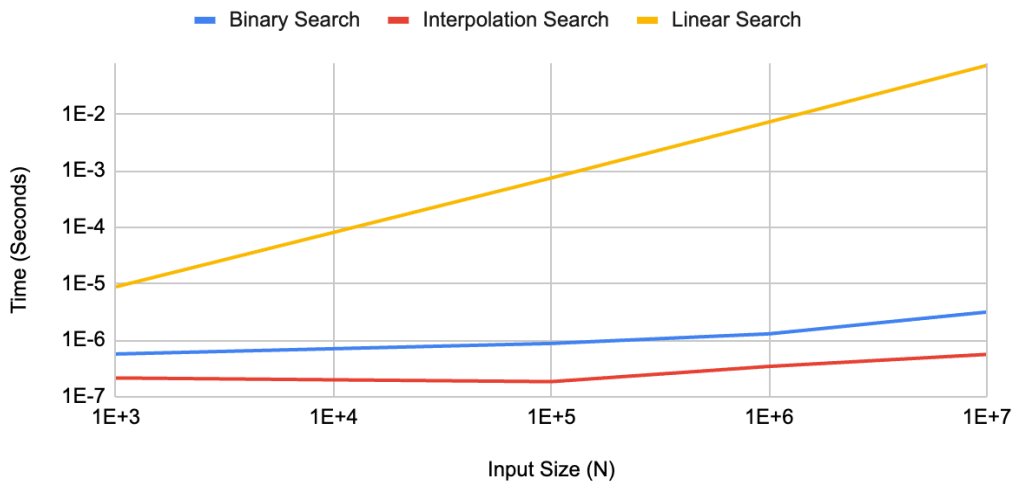


Figure 2. Graph created through Google Sheets using “Linux_search_results.csv”.

Displays minimum runtime comparisons across three different search algorithms given different input sizes

Linux Specs: Intel Core Ultra 7 265K x 20, 32.0 GiB, Linux 7.0.0-31-generic

Minimum Time calculated by calculating average trial times divided by sample size, and identifying smallest value

Search Algorithms Run Time Comparisons on Macbook Pro(over 3 trials, 1000 samples in each trial)

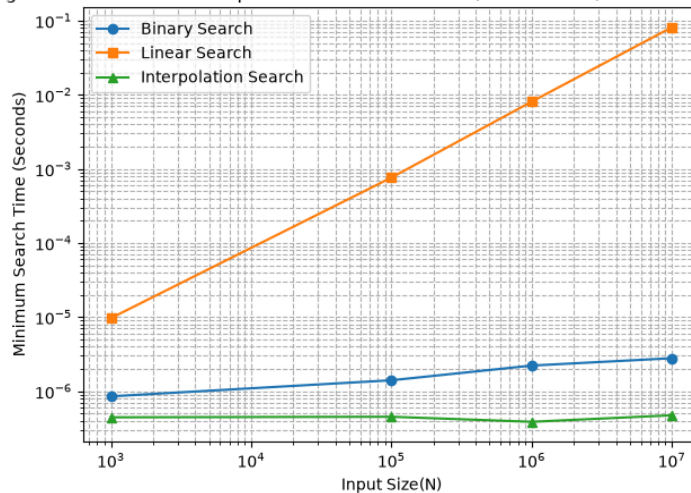


Figure 3. Graph created through matplotlib in timing.ipynb using “macbookPro_search_results.csv”

Displays minimum runtime comparisons across three different search algorithms given different input sizes

Macbook Pro Specs: Apple M3 Pro, 18 GB, MacOS Sequoia Version 15.6.1

Minimum Time calculated by calculating average trial times divided by sample size, and using smallest value

Search Algorithms Run Time Comparisons on Linux(over 3 trials, 1000 samples in each trial)

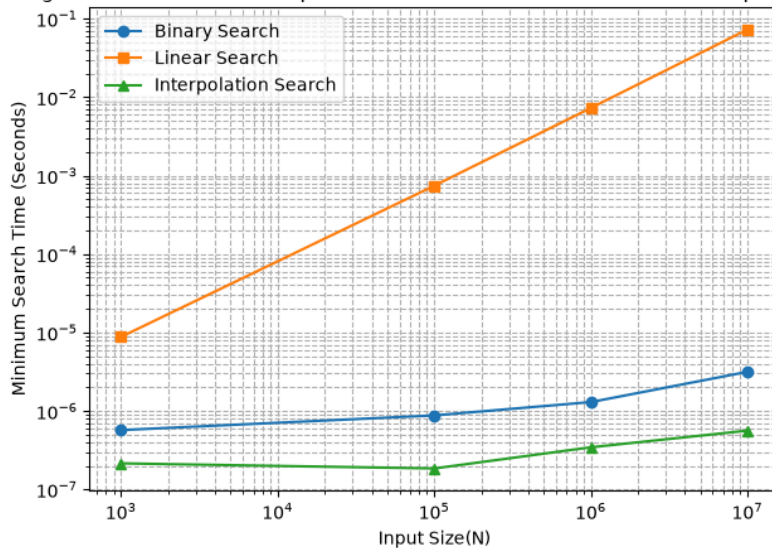


Figure 4. Graph created through matplotlib in timing.ipynb using “Linux_search_results.csv”

Displays minimum runtime comparisons across three different search algorithms given different input sizes

Linux Specs: Intel Core Ultra 7 265K x 20, 32.0 GiB, Linux 7.0.0-31-generic

Minimum Time calculated by calculating average trial times divided by sample size, and identifying smallest value

<i>Input Size (N)</i>	Binary Search(s)	Interpolation Search (s)	Linear Search (s)
1000	5.71E-07	2.15E-07	8.80E-06
100000	8.78E-07	1.85E-07	0.000742667224
1000000	1.30E-06	3.45E-07	0.007274742267
10000000	3.16E-06	5.62E-07	0.07284436091

Table 1. Table created through Google Sheets using “Linux_search_results.csv”

Displays recorded minimum runtime values on Linux machine for three algorithms with varying input sizes

Linux Specs: Intel Core Ultra 7 265K x 20, 32.0 GiB, Linux 7.0.0-31-generic

Minimum Time calculated by calculating average trial times divided by sample size, and identifying smallest value

<i>Input Size (N)</i>	Binary Search (s)	Interpolation Search (s)	Linear Search (s)
1000	8.52E-07	4.44E-07	9.69E-06
100000	1.40E-06	4.52E-07	0.000764104
1000000	2.22E-06	3.86E-07	0.008085662
10000000	2.79E-06	4.73E-07	0.081685243

Table w. Table created through Google Sheets using “macbookPro_search_results.csv”

Displays recorded minimum runtime values on Macbook Pro for three algorithms with varying input sizes

Macbook Pro Specs: Apple M3 Pro, 18 GB, MacOS Sequoia Version 15.6.1

Minimum Time calculated by calculating average trial times divided by sample size, and using smallest value